

RESEARCH ARTICLE

Modular Organization of Rba50 Reveals Distinct Domains for RNAPII Assembly and Interaction Network Formation

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ABSTRACT

RNA polymerase II (RNAPII) is a conserved 12-subunit enzyme essential for eukaryotic transcription. Although the structure and biological functions of RNAPII are well-defined, the mechanisms by which its subunits are assembled into a functional complex remain only partially understood. Several RNAPII assembly factors have been identified, but the molecular principles by which they cooperate during polymerase biogenesis remain unclear. Rba50 is an essential, conserved RNAPII assembly factor implicated in multiple stages of polymerase biogenesis, yet its mode of action remains unknown. Here, we combine structural prediction, targeted interaction assays, and functional analyses to reveal a modular organization of Rba50 with separable activities. Rba50 comprises two functionally distinct but cooperative regions: a structured C-terminal module that provides RNAPII assembly-associated activity and an intrinsically disordered N-terminal module that functions as a multivalent interaction hub. We map a direct interaction between the assembly factor Npa3 and the first N-terminal disordered segment, residues 1–89, and show that the N-terminal module expands the Rba50 interaction network. Functional assays indicate that the C-terminal region rescues specific *rba50-3* phenotypes and supports proliferative recovery under permissive plasmid-shuffle conditions, whereas robust Rba50 function also requires determinants outside the C-terminal region. We further identify a *CRM1*-dependent leucine-rich nuclear export signal within residues 239–249 that limits nuclear accumulation of Rba50 at steady state. Together, our results support a modular architecture in which distinct functional elements within Rba50 contribute to RNAPII biogenesis, interaction-network formation, and nucleocytoplasmic distribution.

1 | Introduction

RNA polymerase II (RNAPII) is the central enzyme of eukaryotic transcription [1–4]. It is composed of 12 subunits that assemble into a conserved and highly structured complex [1, 3, 5, 6]. Although high-resolution structures of RNAPII are available [3], the way through which its subunits are assembled remains only partially understood. Accumulating evidence indicates that RNAPII assembly requires a dedicated set of biogenesis factors, including Rba50, Gpn2, Gpn3, Npa3, Rtr1, and Bud27, which act

at distinct stages of complex formation [7–17]. In both yeast and humans, Rba50 (*RPAP1* in humans) has been shown to interact with several RNAP subunits, including Rpb10, Rpb2, Rpb3, and Rpb11 [14, 18–21], underscoring a conserved functional role.

Previous studies from our laboratory and others have outlined a working model in which Rba50 cooperates with Gpn2 to promote formation of the Rpb3 subcomplex (Rpb3–Rpb10–Rpb11–Rpb12) [7, 22]. At later stages, Rba50 contributes to the incorporation of Rpb2 through coordinated action with Gpn2,

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Npa3, and Gpn3 [14, 17]. Similar principles apply to RNAPIII biogenesis, where Rba50 participates in the assembly of the Rpc40 subcomplex and in the loading of the Rpc128 module [14], highlighting that the underlying mechanism is evolutionarily conserved across the RNAP families. Loss of Rba50 disrupts proper nuclear import and assembly of RNAPII subunits, leading to cytoplasmic accumulation and impaired transcriptional activity [7, 14].

Despite its essential role, the molecular organization of Rba50 and the basis by which it dynamically engages multiple partners remain poorly defined. Mechanistic studies have traditionally focused on folded domains, yet the structure of Rba50 has not been experimentally elucidated. In this study, we combined structural prediction, targeted interaction assays, and functional analyses to reveal a previously unrecognized modular organization of Rba50.

2 | Materials and Methods

2.1 | Yeast Strains and Basic Methods

All strains used in this study were derived from *Saccharomyces cerevisiae* background W303-1a [23] and are listed in Table 1. Strains were generated using standard yeast genetics techniques as previously described [7, 14].

C-terminally GFP-tagged *RBA50* alleles were generated at the endogenous genomic locus by integration of PCR-amplified cassettes derived from the Yeast GFP Clone Collection genomic DNA. The integration cassette contained approximately 1000bp of homology flanking the *RBA50* stop codon. *RBA50* alleles carrying a C-terminal NLS-GFP tag were similarly generated by genomic integration of PCR-amplified cassettes derived from the pRS306-*RBA50*-NLS-GFP:*URA3* plasmid, using primer pairs containing 60bp of homology to sequences flanking the *RBA50* stop codon.

To generate the YDB526 strain, the single-copy plasmid pRS316-*RBA50*-13myc:*URA3* was first transformed into the W303-1a strain. The endogenous *RBA50* locus was then deleted and replaced with the nutritional selection marker *HIS3*. Plasmids pRS425-*LEU2* or pRS315-*LEU2* carrying Rba50-GFP, Rba50 (C-ter₂₀₁₋₄₃₉)-GFP, Rba50 (N-ter₁₋₂₀₀)-GFP, Rba50 (C-ter₂₆₅₋₄₃₉)-GFP, Rba50 (C-ter₂₀₁₋₄₃₉)-NLS-GFP, or Rba50 (N-ter₁₋₂₀₀)-NLS-GFP were subsequently transformed into the YDB526 strain. Finally, counterselection on 5-fluoroorotic acid (5-FOA) was used to eliminate the pRS316-*RBA50*-13myc:*URA3* plasmid, yielding strains expressing only the indicated Rba50 variants from pRS425- or pRS315-based plasmids.

Plasmids pRS425, pRS424, pRS315, and pRS314 were used for multicopy or single-copy expression of genomic DNA fragments corresponding to the different open reading frames using the AFEAP cloning method.

Yeast cells were grown in YPD medium unless otherwise indicated. Strains carrying pRS425- or pRS424-based multicopy plasmids were grown in synthetic dropout (SD) medium lacking leucine, whereas strains carrying pRS315- or pRS314-based

TABLE 1 | Yeast strains in this study.

Strain	Relevant genotype	Source
W303-1a	MATa <i>ade2-1 ura3-1 his3-11,15 trp1-1 leu2-3, 112can1-100</i>	Lab stock
YFZ77	W303-1a <i>rba50Δ::rba50-S288P D293V I344R-13myc:URA3</i>	Lab stock
YFZ84	W303-1a <i>RBA50-GFP:HIS3MX6</i>	Lab stock
YFZ96	W303-1a <i>rba50Δ::rba50-S288P D293V I344R-13myc:URA3 RPB1-GFP:HIS3MX6</i>	Lab stock
YXQ19	W303-1a <i>NPA3-13myc:URA3</i>	Lab stock
YDB170	W303-1a <i>RBA50-NLS-GFP:HIS3MX6</i>	This study
YDB526	W303-1a <i>rba50Δ::HIS3 pRS316-RBA50-13myc:URA3</i>	This study
YDB534	W303-1a <i>rba50Δ::HIS3 pRS315-RBA50-GFP:LEU2</i>	This study
YDB535	W303-1a <i>rba50Δ::HIS3 pRS425-RBA50-GFP:LEU2</i>	This study
YDB536	W303-1a <i>rba50Δ::HIS3 pRS315-RBA50 (C-ter₂₀₁₋₄₃₉)-GFP:LEU2</i>	This study
YDB537	W303-1a <i>rba50Δ::HIS3 pRS315-RBA50 (N-ter₁₋₂₀₀)-GFP:LEU2</i>	This study
YDB538	W303-1a <i>rba50Δ::HIS3 pRS425-RBA50 (C-ter₂₀₁₋₄₃₉)-GFP:LEU2</i>	This study
YDB539	W303-1a <i>rba50Δ::HIS3 pRS425-RBA50 (N-ter₁₋₂₀₀)-GFP:LEU2</i>	This study
YDB540	W303-1a <i>rba50Δ::HIS3 pRS424-RBA50 (C-ter₂₀₁₋₄₃₉)-GFP:TRP1</i>	This study
YDB541	W303-1a <i>rba50Δ::HIS3 pRS314-RBA50 (C-ter₂₀₁₋₄₃₉)-GFP:TRP1</i>	This study
YDB552	W303-1a <i>rba50Δ::HIS3 pRS425-RBA50 (L239A)-GFP:LEU2</i>	This study
YDB553	W303-1a <i>rba50Δ::HIS3 pRS425-RBA50 (L242A)-GFP:LEU2</i>	This study
YDB554	W303-1a <i>crm1Δ::CRM1-T539C:URA3</i>	This study
YDB555	W303-1a <i>crm1Δ::CRM1-T539C:URA3 pRS425-RBA50-GFP:LEU2</i>	This study
YDB556	W303-1a <i>rba50Δ::HIS3 pRS425-RBA50 (C-ter₂₀₁₋₄₃₉)-GFP:LEU2 pRS424-RBA50 (N-ter₁₋₂₀₀)-13myc:TRP1</i>	This study
YDB557	W303-1a <i>rba50Δ::HIS3 pRS424-RBA50 (N-ter₁₋₂₀₀)-13myc:TRP1</i>	This study

single-copy plasmids were grown in SD medium lacking leucine or tryptophan, respectively. YPD and SD media were prepared as previously described [7].

2.2 | Serial Dilution Assay, Protein Extraction, Co-Immunoprecipitation, and Immunoblotting

Serial dilution growth assays, trichloroacetic acid (TCA) whole-cell protein extraction, co-immunoprecipitation, and Western blotting were performed as described previously [7]. Primary antibodies used in this study included anti-Myc (9E10; monoclonal mouse hybridoma supernatant) and anti-GFP (mouse monoclonal GF28R; Thermo Fisher Scientific).

2.3 | GST Pull-Down Assays

GST-Rba50, GST-Rba50 ($\Delta 1-89$), and GST-Rba50 (N-ter₁₋₈₉) were expressed in *E. coli* BL21 cells and purified by affinity chromatography using glutathione-Sepharose beads according to the manufacturer's protocol (GE Healthcare). Pull-down assays were performed to assess physical interactions between Rba50 (or deletion derivatives) and Npa3. Native protein extracts from the indicated yeast strain (YXQ19, Npa3-13myc) were incubated with GST-Rba50-glutathione-agarose beads or GST-glutathione-agarose beads at 4°C for 1 h. Beads were washed three times, and bound proteins were eluted by boiling in Laemmli sample buffer, resolved by SDS-PAGE, and analyzed by Western blotting using anti-Myc antibodies to detect Npa3-13myc.

2.4 | Fluorescence Microscopy

For analysis of Rpb1-GFP localization in the *rba50-3* background, cells carrying pRS425-based constructs were grown in SD medium lacking leucine at 30°C and then shifted to 34°C for 2 h prior to imaging. For analysis of Rba50-GFP and Rba50-NLS-GFP localization, cultures were grown to logarithmic phase in YPD medium before fluorescence microscopy. For localization analysis of Rba50-GFP, Rba50 (C-ter₂₀₁₋₄₃₉)-GFP, Rba50 (N-ter₁₋₂₀₀)-GFP, Rba50 (C-ter₂₆₅₋₄₃₉)-GFP, Rba50 (C-ter₂₀₁₋₄₃₉)-NLS-GFP, Rba50 (N-ter₁₋₂₀₀)-NLS-GFP, and Rba50 ($\Delta 201-265$)-GFP, the corresponding constructs were cloned into pRS425 and expressed in wild-type cells grown in SD medium lacking leucine at 30°C. For plasmid-shuffle localization experiments, *rba50Δ* cells maintained by pRS316-*RBA50*-13myc were transformed with the indicated *LEU2* plasmids and imaged after counterselection on 5-FOA. For Crm1 inhibition, *crm1-T539C* cells expressing pRS425-*RBA50*-GFP were grown to logarithmic phase and treated with 100 ng/ml leptomycin B for 2 h before imaging. Nuclei were counterstained with DAPI. Images were acquired using a DeltaVision Elite microscope (Applied Precision Inc., Mississauga, ON, Canada) and processed with Volocity software.

3 | Results

3.1 | Structural Prediction Unveils an Unexpected Link Between Rba50 and Nucleocytoplasmic Transport

Rba50 and its human homolog RPAP1 are key biogenesis factors required for the assembly of RNA polymerase II (RNAPII) [7, 14, 17, 21, 24, 25]. Previous work from our group established

that Rba50 cooperates with Gpn2 to promote assembly of the Rpb3–Rpb12 module and subsequently contributes to the incorporation of Rpb2 [7, 14]. Rba50 also participates in analogous steps during RNAPIII assembly [14]. However, despite these insights, the molecular basis by which Rba50 performs its multiple functions has remained incompletely understood.

To explore potential structural determinants of Rba50 function, we examined the AlphaFold model of the budding yeast Rba50 protein (AF-Q04418-F1-model_v2) using the Dali server, which compares predicted three-dimensional protein structures to infer possible functional analogies [26]. The analysis revealed that Rba50 shares structural features with numerous proteins involved in different forms of intracellular transport, including nucleocytoplasmic transport, vesicle trafficking, Golgi, and endosomal transport (Figure 1A). Notably, the highest similarity score was obtained with the predicted structure of the importin Kap95 (AF-Q06142-F1-model_v4). Structural superposition in PyMOL showed that a continuous region of Rba50 aligns with a corresponding region in Kap95 (Figure 1B), and sequence extraction from these structurally overlapping segments revealed detectable local similarity (Figure 1C). Consistent with the prevalence of HEAT- and ARM-repeat-based solenoid architectures among transport-related proteins, including importins [27], these observations raised the possibility that Rba50 may contain regulatory elements controlling its nucleocytoplasmic localization. In line with this idea, the Rba50 metazoan homolog *RPAP1/IYO* shows condition- and development-dependent nuclear accumulation in mammals and plants [24, 28]. Prompted by these considerations, we next asked whether Rba50 contains internal elements that regulate its nucleocytoplasmic distribution.

3.2 | A CRM1-Dependent Nuclear Export Signal Controls Rba50 Nucleocytoplasmic Shuttling

To explore whether Rba50 contains determinants of subcellular localization, we fused an SV40-derived NLS and a GFP tag to the C-terminus of the endogenous *RBA50* locus (Figure 2A). The NLS fusion did not alter cell viability or the fluorescence pattern of Rba50, which exhibited a diffuse signal throughout both the nucleus and cytoplasm (Figure 2A). This result is compatible with Rba50 carrying a motif counteracting forced nuclear enrichment.

To identify such a region, we split Rba50 into an N-terminal 1–200 fragment and a C-terminal 201–439 fragment. These boundaries were selected on the basis of AlphaFold structural predictions and disorder analysis, which indicate a transition from an IDR-rich N-terminal region to a more structured C-terminal region near residue 200. Both fragments were fused individually to GFP and to an SV40 NLS (Figure 2B). Expression of the fragments from a multicopy vector in cells carrying endogenous *RBA50* showed that the N-terminal 1–200 fragment accumulated strongly in the nucleus when fused to an NLS. In contrast, the C-terminal 201–439 fragment was less responsive to NLS addition under these conditions (Figure 2C), suggesting that a localization determinant might reside within the C-terminal segment.

We next further truncated the C-terminal fragment at residue 264 (C-terminal 265–439 fragment). This shortened fragment

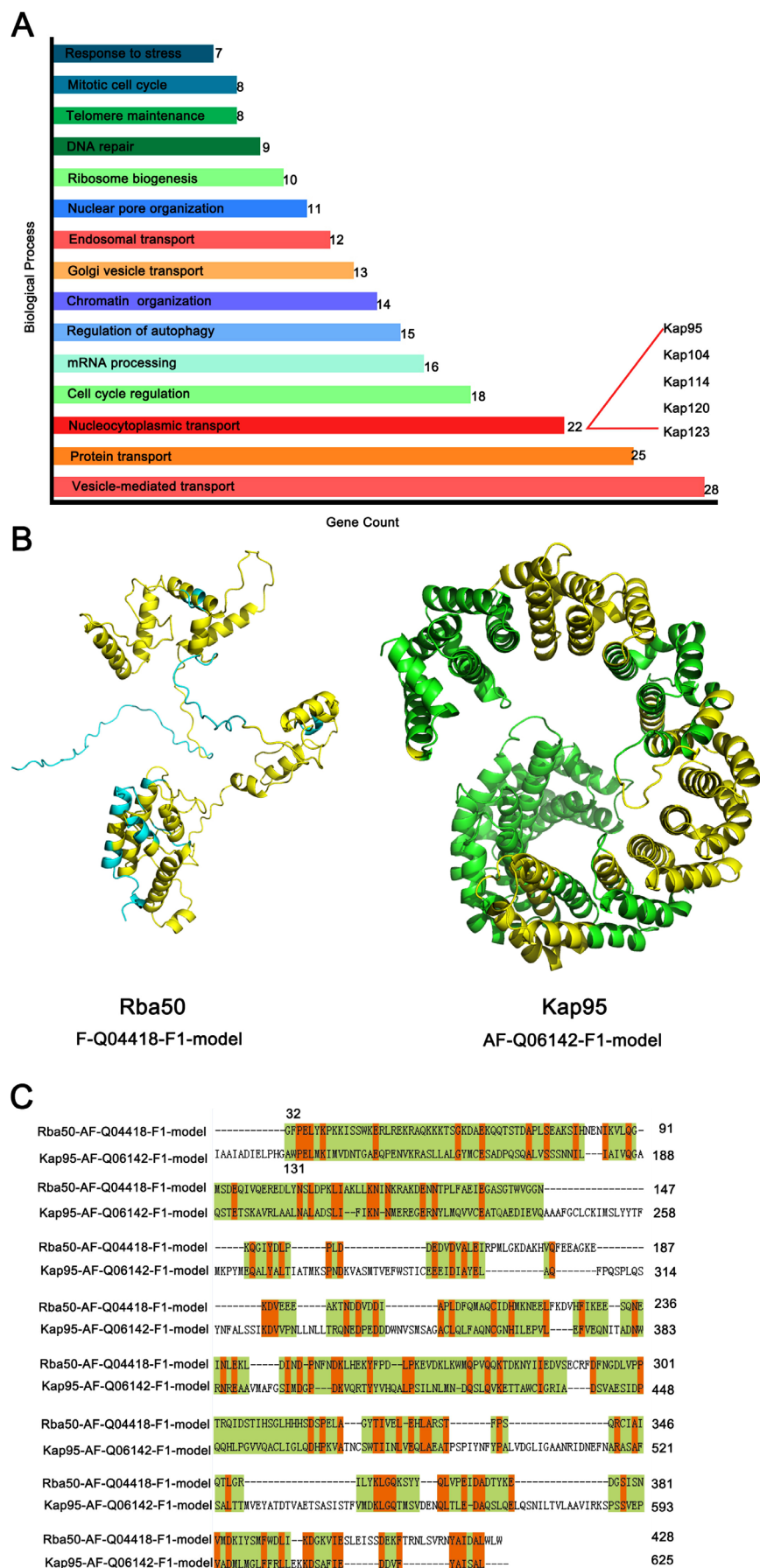


FIGURE 1 | Legend on next page.

FIGURE 1 | Structural analysis and functional inference of Rba50. (A) Structural similarity analysis of Rba50 was performed using the Dali Server with the AlphaFold-predicted Rba50 structure as input. Proteins with a Z-score > 3 were identified and subjected to Gene Ontology biological process enrichment analysis. The top enriched biological process categories are shown, with gene counts indicated. (B) Structural alignment of the AlphaFold-predicted models of Rba50 (AF-Q04418-F1-model_v2) and Kap95 (AF-Q06142-F1-model_v4) performed using PyMOL. Structurally aligned regions are highlighted. (C) Sequence alignment of structurally corresponding regions between Rba50 and Kap95, highlighting conserved or similar residues within aligned segments.

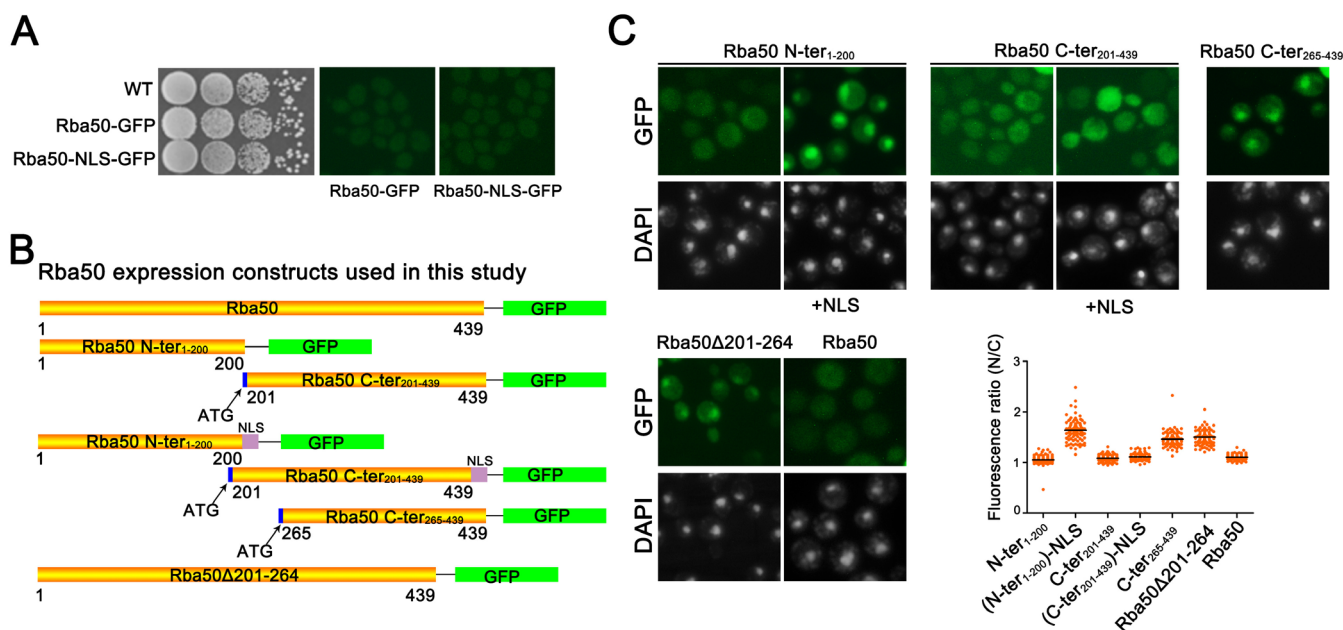


FIGURE 2 | Peptide segment 201–264 regulates the nucleocytoplasmic localization of Rba50. (A) Left panel: Ten-fold serial dilutions of exponentially growing wild-type cells (W303-1a), Rba50-GFP (YFZ84), and Rba50-NLS-GFP (YDB170) spotted on YPD plates and incubated at 30°C. Right panel: Subcellular localization of Rba50-GFP and Rba50-NLS-GFP analyzed by fluorescence microscopy. (B) Summary of the RBA50 expression constructs used in this study. Yellow bars indicate Rba50 fragments, green indicates GFP, purple indicates the SV40 nuclear localization signal (NLS), and blue indicates the position at which an ATG start codon was introduced. (C) Subcellular localization of Rba50 truncation and deletion variants. Wild-type cells carrying the indicated GFP-tagged constructs were grown to logarithmic phase and analyzed by fluorescence microscopy. Nuclei were visualized by DAPI staining. Strip plot representation of the nuclear-to-cytoplasmic fluorescence ratios for the indicated strains; each dot represents one cell ($n = 100$ for each strain), and horizontal black bars show the mean value.

displayed robust nuclear accumulation, suggesting that the determinant limiting nuclear enrichment may reside, at least in part, within the 201–264 sequence (Figure 2C, right panel).

To directly evaluate the contribution of this region to subcellular localization, we deleted residues 201–264 in full-length Rba50 fused to GFP. Removal of this segment resulted in strong nuclear accumulation, resembling the behavior of the C-terminal 265–439 fragment (Figure 2C). These initial mapping experiments, therefore, implicated residues 201–264 in the regulation of Rba50 nucleocytoplasmic distribution.

The localization results were compatible with residues 201–264 containing a regulatory sequence that restricts nuclear enrichment of Rba50. Sequence inspection of this region revealed two short hydrophobic stretches similar to leucine-rich nuclear export signals (NESs). Specifically, residues 222–229 (LFKDVHFI) and 239–249 (LEKLDINDPNF) resemble the canonical NES consensus Φ -X₂₋₃- Φ -X₂₋₃- Φ -X- Φ , where Φ represents a hydrophobic residue such as leucine, isoleucine, valine, or phenylalanine (Figure 3A).

To test whether these sequences function as NES elements, we substituted conserved hydrophobic residues within each motif with alanine. Specifically, the L222A, L239A, and L242A mutations were introduced into full-length Rba50 fused to GFP, and the subcellular localization of the resulting proteins was examined in vivo. Whereas the L222A mutant displayed a diffuse distribution throughout the nucleus and cytoplasm, the L239A and L242A mutants exhibited pronounced nuclear accumulation (Figure 3B), indicating that disruption of either of these two residues impairs nuclear exclusion.

To determine whether Rba50 nuclear export depends on the classical CRM1-exportin pathway, we examined Rba50-GFP localization in a *crm1-T539C* strain, which is sensitive to leptomycin B (LMB). Treatment with 100 ng/mL LMB for 2 h resulted in clear nuclear accumulation of Rba50-GFP compared with untreated cells (Figure 3C). Together, the L239A and L242A localization phenotype and the LMB sensitivity demonstrate that residues 239–249 contain a functional CRM1-dependent leucine-rich NES that restricts Rba50 nuclear accumulation.

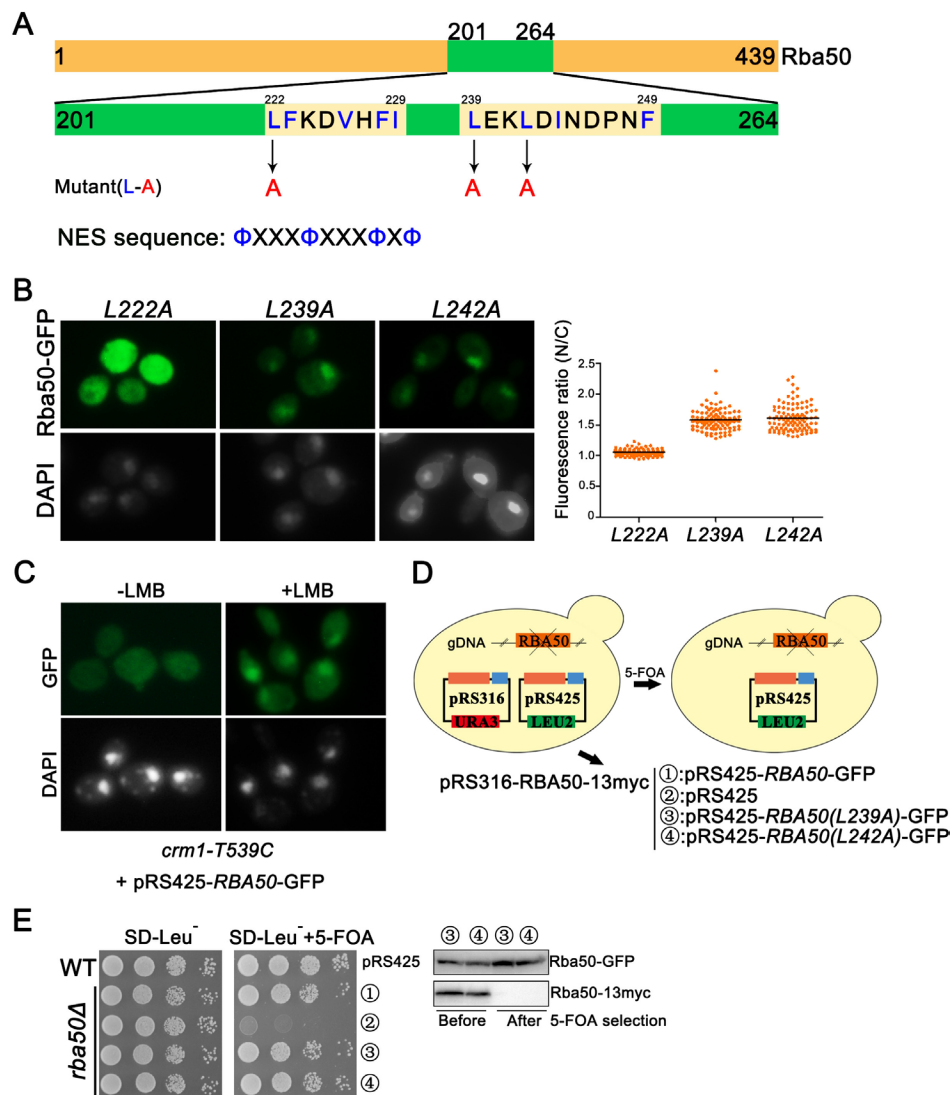


FIGURE 3 | The 239–249 sequence contains a functional CRM1-dependent NES. (A) Schematic representation of the hydrophobic motifs within residues 201–264 and the positions of leucine-to-alanine substitutions. (B) Subcellular localization of Rba50 leucine point mutants. Wild-type cells carrying the indicated plasmids were grown to logarithmic phase and analyzed by fluorescence microscopy. Nuclei were visualized by DAPI staining. Strip plot representation of the nuclear-to-cytoplasmic fluorescence ratios for the indicated strains; each dot represents one cell ($n = 100$ for each strain), and horizontal black bars show the mean value. (C) Rba50-GFP localization in *crm1-T539C* cells in the absence or presence of leptomycin B (LMB, 100 ng/mL). Nuclei were visualized by DAPI staining. (D) Schematic of the plasmid-shuffling strategy used to assess the functional relevance of residues 201–264. Loss of the URA3-marked pRS316-RBA50-13myc plasmid under 5-FOA selection is indicated. (E) Left panel: Ten-fold serial dilutions of wild-type and *rba50Δ* mutant cells carrying the indicated mutant constructs were spotted on SD-Leu plates with or without 5-FOA and incubated at 30°C for 7 days. Right panel: Western blot analysis of whole cell extracts before and after 5-FOA selection was performed to assess the presence of the indicated Rba50 variants.

We next asked whether the disruption of this NES compromises the essential function of Rba50. To this end, we used a plasmid-shuffle assay in *rba50Δ* cells maintained by a URA3-RBA50 plasmid (Figure 3D). Cells transformed with plasmids expressing Rba50-GFP, Rba50 (L239A)-GFP, or Rba50 (L242A)-GFP were plated on 5-FOA to counterselect the URA3-RBA50 plasmid. Cells relying exclusively on either NES mutant remained viable, and Western blot analysis confirmed loss of Rba50-13Myc after counterselection and expression of the GFP-tagged Rba50 variants (Figure 3E). Thus, although mutation of L239 or L242 alters steady-state localization, NES disruption does not abolish the essential function of Rba50 under the conditions tested.

Finally, because endogenous wild-type Rba50 could potentially interfere with the localization of the plasmid-encoded fragments, we examined fragment localization after plasmid shuffling in *rba50Δ* cells maintained by a URA3-RBA50 plasmid (Figure S1). Under these conditions, the NLS-tagged N-terminal fragment showed nuclear enrichment, whereas full-length Rba50-NLS-GFP and the NLS-tagged C-terminal fragment remained more broadly distributed, in agreement with the results in Figure 2. Thus, the localization pattern observed for the NLS-tagged viable constructs is retained in the absence of endogenous Rba50. Importantly, these cells are not uncompensated *rba50Δ* null cells, but *rba50Δ* cells expressing plasmid-borne Rba50 fragments. Thus, recovery after 5-FOA counterselection indicates that either isolated fragment

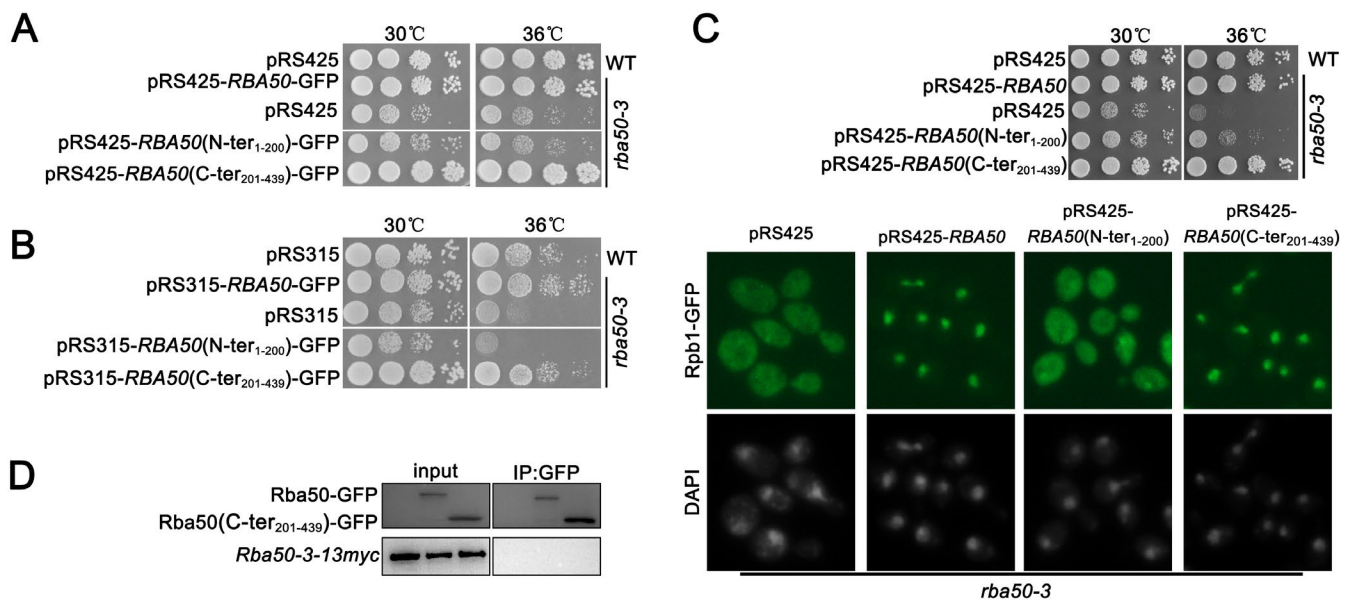


FIGURE 4 | The C-terminal region of Rba50 rescues *rba50-3* mutant phenotypes. (A) Ten-fold serial dilutions of exponentially growing *rba50-3* mutant cells (YFZ77) carrying the indicated multicopy plasmids were spotted on SD-Leu plates and incubated for 4 days at permissive or restrictive temperatures, as indicated. Wild-type cells carrying an empty vector were included as controls. (B) Ten-fold serial dilutions of *rba50-3* mutant cells carrying the indicated single-copy plasmids were analyzed as in (A). (C) Upper panel: Growth analysis of temperature-sensitive *rba50-3* mutant cells (YFZ96) carrying the indicated multicopy constructs. Lower panel: Subcellular localization of the RNAPII subunit Rpb1 in the indicated *RBA50* backgrounds. Cells were first grown to mid-exponential phase at 30°C, then shifted to 34°C for 2 h and fixed. The localization of Rpb1-GFP was analyzed by fluorescence microscopy. (D) Co-immunoprecipitation analysis testing whether Rba50-GFP or Rba50 (C-ter₂₀₁₋₄₃₉)-GFP stably associates with *rba50-3-13myc*. Whole-cell extracts were immunoprecipitated using anti-GFP beads, and precipitated proteins were analyzed by immunoblotting with anti-GFP and anti-Myc antibodies.

retains partial Rba50 activity sufficient to support proliferation under permissive plasmid-shuffle conditions, rather than demonstrating full replacement of intact Rba50. Notably, shorter C-terminal fragments beginning at residues 202 or 203 could not be recovered after counterselection, suggesting that the integrity of the 201–439 fragment boundary is functionally relevant.

3.3 | The C-Terminal Region of Rba50 Restores RNAPII Assembly-Associated Phenotypes in *rba50-3* Mutant Cells

To test whether the N- and C-terminal regions of Rba50 make separable functional contributions, we expressed Rba50 N-terminal and C-terminal fragments from single- or multicopy plasmids in the temperature-sensitive *rba50-3* mutant. The *rba50-3* allele, which exhibits a severe defect in the assembly of RNA polymerase II at a restrictive temperature [7], displays a pronounced growth defect at semi-restrictive temperature (36°C). Both multi- and single-copy Rba50 C-terminal constructs restored growth of *rba50-3* cells at semi-restrictive temperature, whereas the Rba50 N-terminal construct did not (Figure 4A,B).

To investigate whether the rescue of the *rba50-3* growth defect reflected improvement of RNAPII biogenesis, we expressed the untagged Rba50 C-terminal fragment in *rba50-3* cells carrying Rpb1-GFP. Expression of the Rba50 C-terminal fragment corrected the Rpb1-GFP mislocalization observed in *rba50-3* cells (Figure 4C), indicating that the C-terminal region is sufficient to restore this RNAPII assembly-associated readout. In the same

assay, expression of the Rba50 N-terminal fragment failed to restore nuclear localization of Rpb1-GFP, consistent with the major assembly-associated activity residing in the C-terminal region.

The three amino-acid substitutions of the *rba50-3* allele (S288C, D293V, and I344R) are located within the C-terminal region. Because the wild-type C-terminal sequence rescues the *rba50-3* mutant, one possible mechanism is that the fragment cooperates with the mutant full-length protein through dimeric or oligomeric interactions. To assess whether such a stable association could be detected, we tested whether Rba50-GFP or Rba50 (C-ter₂₀₁₋₄₃₉)-GFP co-purifies with *rba50-3-13myc* from whole-cell extracts. However, under the extraction conditions used, we did not detect a stable association between these proteins (Figure 4D).

Together, these results indicate that, in the *rba50-3* background, the C-terminal region contains the assembly-associated activity required to restore RNAPII assembly-associated phenotypes in vivo. We do not infer from these data that the C-terminal region alone directly mediates every biochemical step of RNAPII assembly; rather, the results identify it as the minimal region sufficient to rescue the growth and Rpb1-localization defects assayed here.

3.4 | Rba50 Operates as a Modular Protein Whose N- and C-Terminal Regions Make Distinct Contributions to Rba50 Function

We next examined the functional contributions of the two modules to Rba50 function in vivo in the absence of endogenous

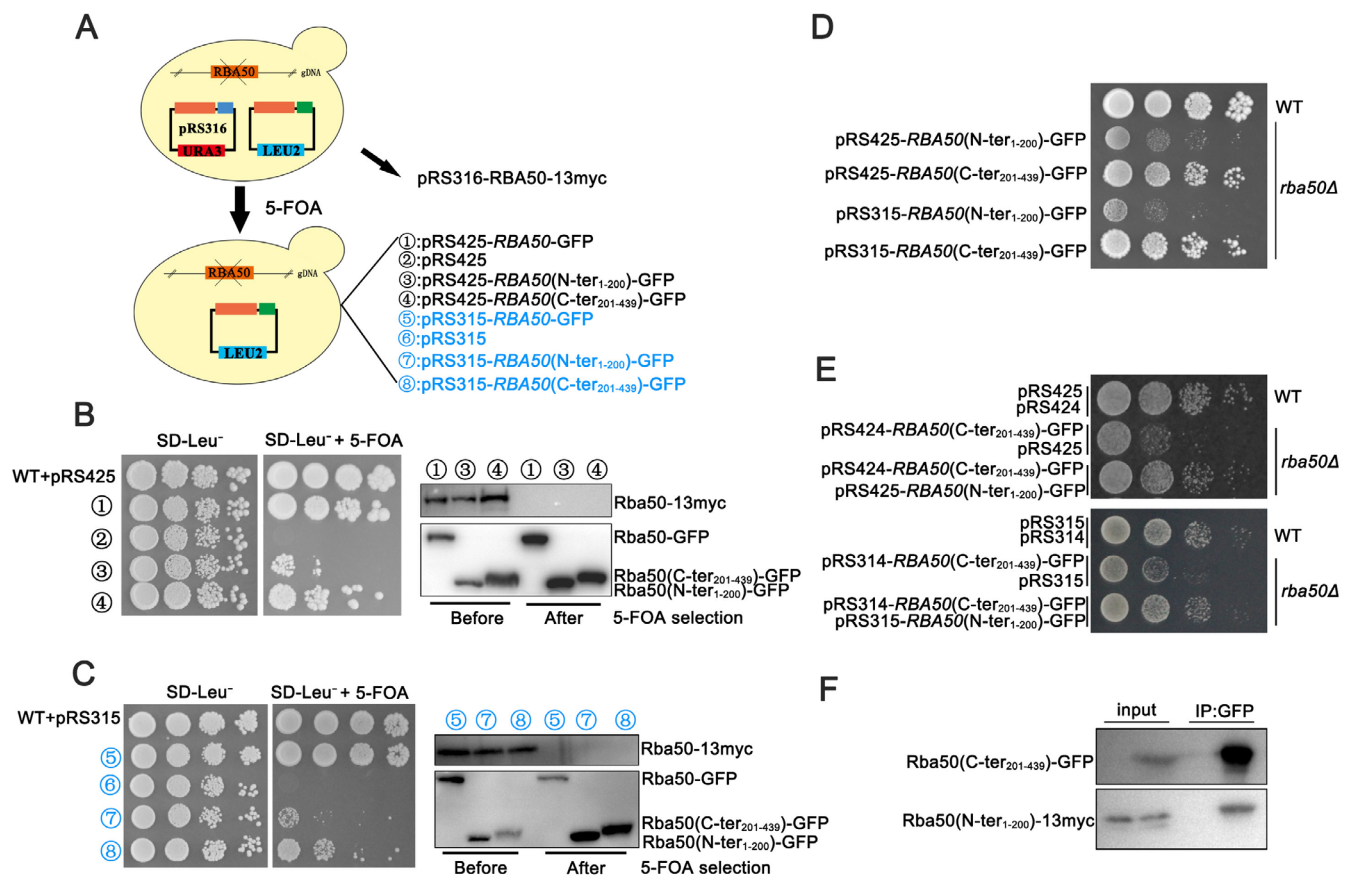


FIGURE 5 | The N-terminal and C-terminal regions of Rba50 make distinct contributions to Rba50 function. (A) Schematic representation of the plasmid-shuffling strategy used to assess the ability of Rba50 fragments to support viability following loss of the URA3-marked pRS316-RBA50-13myc plasmid under 5-fluoroorotic acid (5-FOA) counterselection. (B, C) Left panels: Ten-fold serial dilution assays of wild-type (W303-1a) and *rba50Δ* cells carrying multicopy (B) or single-copy (C) plasmids expressing the indicated Rba50 fragments, spotted on SD-Leu plates with or without 5-FOA and incubated at 30°C for 7 days. Right panels: Immunoblot analysis of TCA whole-cell extracts confirms expression of the indicated Rba50 constructs before and after 5-FOA selection. (D) Growth assays of *rba50Δ* cells expressing either the N-terminal or the C-terminal Rba50 fragments from multicopy or single-copy plasmids following 5-FOA selection, assayed by 10-fold serial dilution on YPD plates incubated at 30°C for 6 days. (E) Growth assays comparing cells expressing C-terminal Rba50 fragments alone or together with N-terminal Rba50 fragments, using the indicated plasmid combinations, following 5-FOA selection. Cells were spotted on SD-Leu-Trp plates and incubated at 30°C for 3 days. (F) Co-immunoprecipitation analysis of Rba50 (C-ter₂₀₁₋₄₃₉)-GFP and Rba50 (N-ter₁₋₂₀₀)-13myc expressed in *rba50Δ* cells. Whole-cell extracts were immunoprecipitated using anti-GFP beads, and precipitated proteins were analyzed by immunoblotting with anti-GFP and anti-Myc antibodies. Input lanes correspond to extracts expressing the indicated constructs before immunoprecipitation.

Rba50. To this end, we used the plasmid-shuffle strategy in *rba50Δ* cells shown before, in which viability is initially maintained by a single-copy plasmid expressing Rba50-13Myc. Cells were transformed with single- or multicopy plasmids expressing either the Rba50 C-terminal or the N-terminal module. Finally, the plasmid carrying the full-length Rba50 was counter-selected on 5-FOA (Figure 5A). Western blot analysis confirmed the absence of full-length Rba50-13Myc (Figure 5B,C).

Under these conditions, cells expressing only the C-terminal or the N-terminal region of Rba50 remained viable but exhibited reduced growth relative to wild type (Figure 5B,C), with the C-terminal fragment providing stronger support.

Because residual wild-type Rba50 protein may persist in cells immediately plated onto 5-FOA, we performed parallel experiments using cultures that had been pre-counterselected prior to plating. Under these conditions, cells expressing the

C-terminal fragment alone were able to form colonies after prolonged incubation, whereas cells expressing only the N-terminal fragment grew poorly (Figure 5D). We then compared cells expressing the C-terminal fragment alone with cells co-expressing the C-terminal and N-terminal fragments under a shorter incubation time. Under these conditions, co-expression of the two fragments improved growth relative to the C-terminal fragment alone and approached the growth of cells expressing full-length Rba50 (Figure 5E). Thus, the C-terminal region retains activity sufficient to support slow proliferative recovery, whereas efficient growth under more stringent assay conditions is enhanced by the N-terminal region.

Consistent with a model in which the two regions can cooperate within the same functional complex, co-immunoprecipitation experiments revealed a robust association between the N-terminal and C-terminal fragments in vivo (Figure 5F).

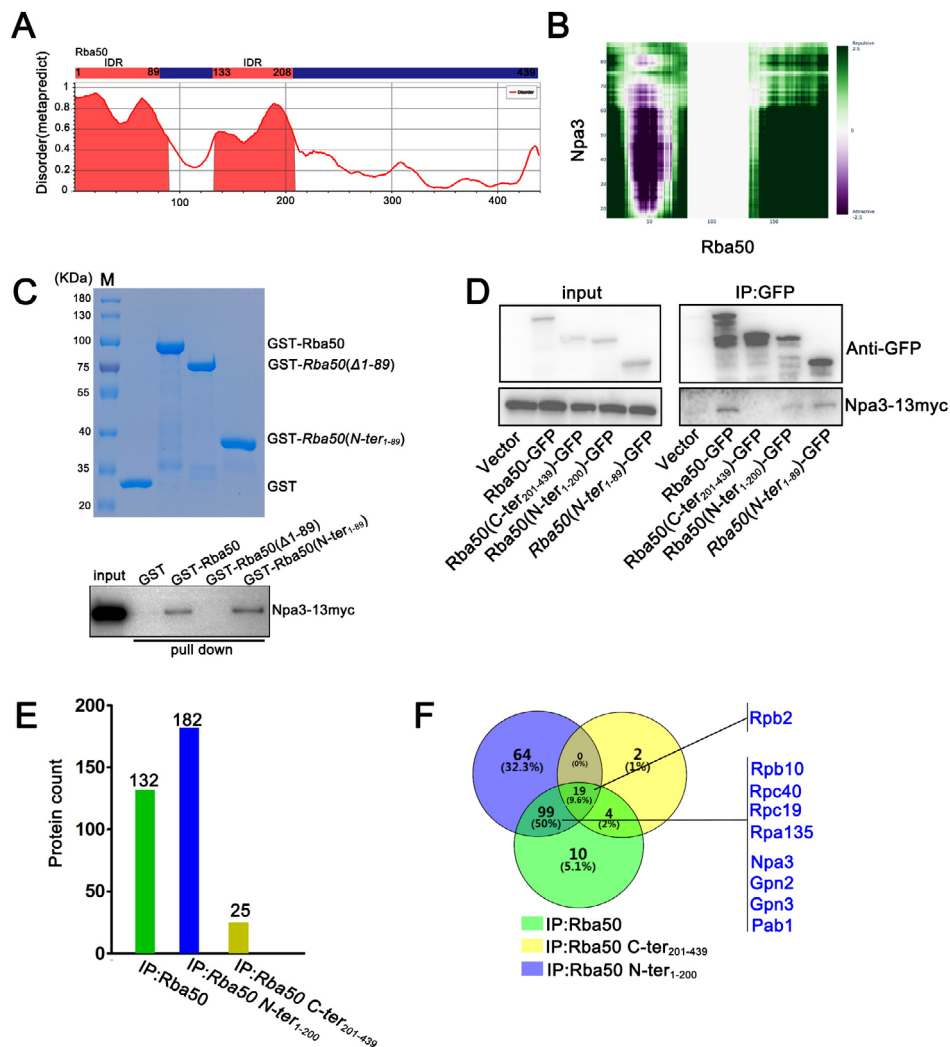


FIGURE 6 | The N-terminal region of Rba50 functions as a multivalent interaction platform. (A) Disorder prediction of Rba50 using Metapredict, highlighting intrinsically disordered regions within the N-terminal portion of the protein. (B) Predicted interaction landscape between Rba50 and Npa3 generated using FINCHES, indicating preferential interaction with the N-terminal region of Rba50. (C) Upper panel: Coomassie-stained SDS-PAGE of GST pull-down experiments using purified recombinant GST-Rba50, GST-Rba50 (Δ1-89), and GST-Rba50 (N-ter₁₋₈₉) on whole cell extracts from Npa3-13myc-expressing cells. Lower panel: Western blot with anti-myc antibodies was used to detect Npa3-13myc. (D) Co-immunoprecipitation of GFP-tagged Rba50 variants with Npa3-13myc from cell extracts, analyzed by Western blotting with anti-GFP and anti-myc antibodies. (E) Mass spectrometry analysis of proteins co-purified with full-length Rba50, Rba50 (N-ter₁₋₂₀₀), or Rba50 (C-ter₂₀₁₋₄₃₉) following GFP immunoprecipitation from *rba50Δ* cells. (F) Overlap analysis of interacting proteins identified by mass spectrometry using jvenn, highlighting shared and unique interaction partners of each Rba50 variant.

3.5 | The N-Terminal Region of Rba50 Mediates a Direct Interaction With the Assembly Factor Npa3

Bioinformatic analysis using Metapredict [29] identified two intrinsically disordered regions (IDRs) within the N-terminal half of Rba50 (amino acids 1–89 and 133–208) (Figure 6A). Given that IDRs frequently serve as multivalent interaction platforms [30], we hypothesized that the N-terminal region might mediate interactions with RNAPII assembly factors. Because previous studies identified Npa3 as a partner of Rba50 and Gpn3 during Rpb2 loading [17], we examined whether Npa3 interacts with one of the predicted IDRs.

The FINCHES predictor [31] suggested that an IDR in Npa3 interacts with the first Rba50 IDR (amino acids 1–89) (Figure 6B).

GST pull-downs using bacterially expressed GST-Rba50, GST-Rba50 (N-ter₁₋₈₉), and GST-Rba50 (Δ1-89) confirmed the prediction. Npa3-13Myc from yeast lysates binds to full-length Rba50 and to the Rba50 N-terminal 1–89 fragment, but not to Rba50 lacking amino acids 1–89 (Figure 6C). This result indicates that residues 1–89 of Rba50 are required for binding to Npa3 in vitro.

To validate the interaction in vivo, we expressed full-length and truncated Rba50 variants in Npa3-13Myc cells and performed co-immunoprecipitation assays. Npa3-13Myc co-purified with full-length Rba50, the Rba50 N-terminal fragment, and the Rba50 N-terminal 1–89 fragment (Figure 6D). Together, these data support that the first IDR (amino acids 1–89) in Rba50 mediates interaction with Npa3 both in vitro and in vivo.

3.6 | The N-Terminal Region of Rba50 Serves as a Multivalent Interaction Hub

Because the N-terminal region of Rba50 contains extensive disordered segments (Figure 6A), we asked whether it acts more generally as a scaffold for protein–protein interactions. We performed IP–MS analysis using full-length Rba50, the Rba50 N-terminal module (amino acids 1–200), or the Rba50 C-terminal module (amino acids 201–439), each expressed in cells lacking chromosomal *RBA50*. Comparison of the three datasets revealed that the Rba50 C-terminal module recovered only a small number of interactors relative to full-length Rba50 (Figure 6E,F and Tables S1–S3). In contrast, the Rba50 N-terminal module co-purified with a substantial subset of the interactome detected for full-length Rba50.

To provide biological context for the IP–MS datasets, we analyzed the identified proteins according to functional categories and known RNAP biogenesis associations. The N-terminal interactome contained known RNAP assembly factors, including Gpn2, Gpn3, and Npa3, as well as RNAP-associated subunits shared among polymerase assembly pathways, including Rpb10, Rpc40, Rpc19, and Rpa135. By contrast, the C-terminal dataset was more restricted and included fewer stable interactors, consistent with a role as a compact functional module rather than a broad interaction scaffold. Taken together, these results support a model in which the N-terminal region functions as an IDR-rich interaction hub that expands the Rba50-associated protein network, whereas the C-terminal region provides a more limited but functionally critical assembly-associated activity.

4 | Discussion

In this study, we combined structural prediction, localization analyses, functional genetics, and interaction assays to define a previously unrecognized modular architecture of Rba50 and to dissect how its distinct regions contribute to RNAPII biogenesis. Although Rba50 has long been recognized as a core assembly factor for RNAPII and RNAPIII [7, 14], the molecular principles underlying its diverse functions have remained unclear. By integrating structural predictions and functional and interaction-based analyses, we uncover features of Rba50 that refine the current understanding of how polymerase assembly is coordinated in eukaryotic cells.

Structural comparisons using the AlphaFold-predicted model of Rba50 revealed similarity to several transport-related proteins, including importins and factors involved in vesicular or endosomal trafficking. The most prominent similarity was with Kap95, a principal receptor for classical nuclear import [32]. Although structural similarity does not imply functional equivalence, this overlap suggests that Rba50 may share structural features with transport regulators. Consistent with observations in mammals and plants, where *RPAP1/IYO* undergoes regulated nuclear accumulation during differentiation [24, 28], our results extend this observation to budding yeast, suggesting that Rba50 harbors an intrinsic nucleocytoplasmic shuttling capacity.

In this respect, we identified a *CRM1*-dependent leucine-rich NES in Rba50 and showed that two different leucine-to-alanine substitutions, *L239A* and *L242A*, resulted in pronounced nuclear

accumulation of the protein. Pharmacological inhibition of Crm1 using LMB further caused nuclear enrichment of Rba50-GFP, demonstrating that Rba50 nuclear export depends on the classical exportin pathway. Notably, this altered distribution did not compromise cell viability, despite Rba50 being an essential factor for RNAPII assembly, a process proposed to occur in the cytoplasm prior to nuclear import [7, 14]. This apparent discrepancy can be reconciled by considering the temporal and kinetic aspects of Rba50 function rather than its steady-state distribution. Thus, NES disruption may primarily increase nuclear retention without fully abolishing cytosolic availability. Even a small cytosolic pool of Rba50, perhaps derived from *de novo* synthesis, may be sufficient to sustain RNAPII assembly. Future work will aim to determine whether regulated nucleocytoplasmic shuttling of Rba50 becomes functionally important under conditions that impose increased demands on RNA polymerase II biogenesis, such as transcriptional stress.

Our work further shows that Rba50 consists of two functionally separable yet cooperative modules. The N-terminal region, enriched in intrinsically disordered sequences (IDRs), provides the primary interaction surface for assembly factors and polymerase subunits. Consistent with the established role of IDRs as flexible, multivalent interaction platforms, residues 1–89 are required for binding to Npa3, and the broader N-terminal domain co-purifies with a wide array of Rba50-interacting proteins *in vivo*, including associations with GPN-family proteins and shared RNAP subunits such as Rpc40, Rpc19, and Rpb10. The N-terminal region thus likely functions as an interaction hub through which Rba50 engages the RNAPII assembly network.

Interestingly, RNA polymerase I components, including Rpa190 and Rpa49, were also recovered, suggesting a potential intersection between the Rba50 N-terminal interactome and RNA polymerase I-associated proteins, the functional significance of which remains to be explored.

In contrast, the C-terminal region does not serve as a major interaction platform. Instead, it carries substantial activity required to restore RNAPII assembly-associated phenotypes, as it rescues growth defects in *rba50-3* mutant cells, corrects Rpb1 mislocalization, and supports colony formation after prolonged plasmid-shuffle outgrowth under permissive conditions. These observations suggest that important activity required to support the assayed RNAPII biogenesis readouts resides within the C-terminal region, but they should not be interpreted as evidence that the isolated C-terminal region is functionally equivalent to full-length Rba50.

Although the C-terminal region of Rba50 associates with far fewer proteins than the N-terminal region in IP–MS analyses, the stable interactors detected are compatible with its ability to support partial growth recovery and selected RNAPII assembly-associated readouts. Notably, the C-terminal fragment retains association with the RNAPII subunit Rpb2, a key substrate of Rba50-dependent assembly, together with general chaperone and protein quality-control factors such as Hsp70 family members and Cdc48. This limited but functionally relevant interaction set is compatible with a model in which the C-terminal region provides assembly-associated activity. By contrast, the N-terminal IDRs expand the interaction network, thereby

enhancing efficiency, coordination, and robustness of the assembly process in the context of the full-length protein.

The plasmid-shuffle growth assays should, therefore, be interpreted as assays of recovery under selection rather than as direct tests of equivalence to full-length Rba50. This distinction is important because the recovered strains are not uncompensated *rba50Δ* null cells, but cells expressing plasmid-borne Rba50 fragments after loss of the *URA3*-marked full-length *RBA50* plasmid. Thus, recovery after counterselection indicates that these fragments retain activity above a minimal viability threshold under permissive conditions, not that they fully substitute for intact Rba50. Consistent with this interpretation, shorter C-terminal fragments beginning at residues 202 or 203 could not be recovered after counterselection, indicating that fragment-supported recovery is boundary-dependent rather than a nonspecific consequence of expressing any C-terminal segment. Growth recovery can also be influenced by incubation time, plasmid copy number, selection conditions, residual protein during early counterselection, and slow outgrowth of cells that maintain sufficient fragment activity. Accordingly, the prolonged-incubation assay and the shorter assay format report different aspects of fragment-supported recovery: eventual colony formation versus early growth kinetics. Thus, the C-terminal region retains activity sufficient to support recovery under permissive plasmid-shuffle conditions, whereas robust Rba50 function depends on additional determinants present in the intact protein.

This modular organization suggests an evolutionary logic in which a conserved C-terminal region provides a core assembly-associated activity, whereas the flexible and expandable N-terminal IDRs confer adaptability by mediating species-specific or condition-specific interactions. Such an arrangement would allow RNAPII biogenesis to be tuned in response to cellular or environmental demands, balancing structural conservation with regulatory plasticity across eukaryotes. Comparative analyses across eukaryotes may reveal how the balance between structural conservation and IDR-driven plasticity has shaped the control of polymerase biogenesis.

Although our genetic and cell-biological data indicate that the C-terminal region of Rba50 is sufficient to restore selected RNAPII assembly-associated phenotypes, we have not directly measured the formation of the Rpb2–Rpb3 subcomplex in the presence of individual Rba50 fragments. Direct biochemical assays of Rpb2–Rpb3 association will be important in future work to define precisely how each Rba50 module contributes to distinct steps of RNAPII assembly. Future work will also need to define high-resolution structures of Rba50 and its complexes with Gpn2, Npa3, and RNAPII subunits to clarify how its two modules cooperate and how transport-associated elements interface with its assembly role. Together, our results provide a framework linking structural prediction, modular organization, nucleocytoplasmic transport, and IDR biology to explain how Rba50 contributes to RNAPII assembly.

Author Contributions

Conceived and designed the experiments: F.Z., D.Q., and D.X. Performed the experiments: D.X., L.W., X.S., Y.Y., and M.G. Analyzed

the data: F.Z., D.Q., D.X., L.W., X.S., P.L., and M.G. Wrote the paper: F.Z., D.Q., and D.X.

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Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Localization analysis of Rba50 constructs after plasmid shuffling in *rba50Δ* cells. (A) Schematic of the plasmid-shuffle strategy used to eliminate the *URA3*-marked pRS316-*RBA50*-13myc plasmid and analyze plasmid-encoded Rba50 constructs as the sole source of Rba50. The indicated constructs include full-length Rba50-NLS-GFP, empty vector, Rba50 (N-ter_{1–200})-NLS-GFP, Rba50 (C-ter_{201–439})-NLS-GFP, Rba50 (C-ter_{201–439})-GFP, Rba50 (C-ter_{202–439})-GFP, and Rba50 (C-ter_{203–439})-GFP. (B) Growth assay and immunoblot analysis of *rba50Δ* cells carrying the indicated Rba50 constructs before and after 5-FOA counterselection. Cells were spotted on SD or SD + 5-FOA plates and incubated at 30°C. Immunoblots confirm loss of Rba50-13myc and expression of the indicated GFP-tagged constructs after counterselection. (C) Fluorescence microscopy of *rba50Δ* cells expressing full-length Rba50, Rba50 (N-ter_{1–200}), or Rba50 (C-ter_{201–439}), with or without an SV40 NLS. Nuclei were visualized by DAPI staining. Strip plot representation of the nuclear-to-cytoplasmic fluorescence ratios for the indicated strains; each dot represents one cell ($n = 100$ for each strain), and horizontal black bars show the mean value. **Table S1:** IP-MS proteins interacted with Rba50. **Table S2:** IP-MS proteins interacted with Rba50 (N-ter_{1–200}). **Table S3:** IP-MS proteins interacted with Rba50 (C-ter_{201–439}).